

Bayesian multimodeling: model complexity

MIPT

2026

Recap: Model selection problem: Bayesian coherent inference

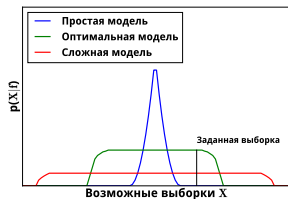
First level: find optimal parameters:

$$\hat{w} = \arg \max \frac{p(\mathcal{D}|\mathbf{w})p(\mathbf{w}|\mathbf{h})}{p(\mathcal{D}|\mathbf{h})},$$

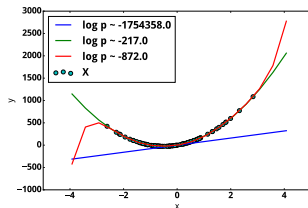
Second level: find model, that gives optimal Evidence.

“Evidence”:

$$p(\mathcal{D}|\mathbf{h}) = \int_{\mathbf{w}} p(\mathcal{D}|\mathbf{w})p(\mathbf{w}|\mathbf{h})d\mathbf{w}.$$



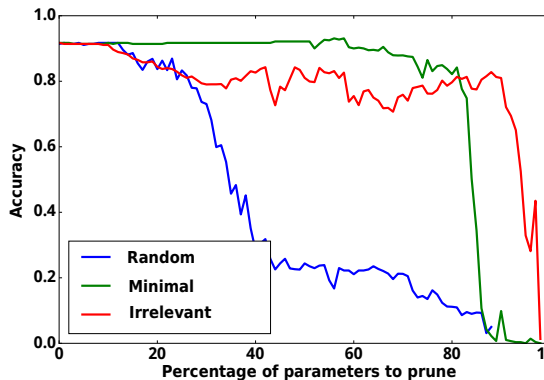
Model selection scheme



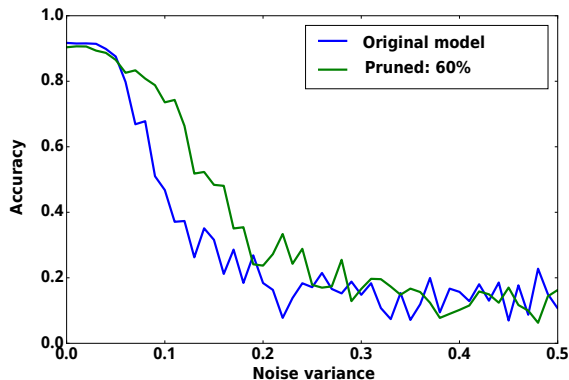
Example: polynomial regression

Model structure selection challenge

Data likelihood does not change with removing redundant parameters.



Redundancy of model parameters



Model robustness

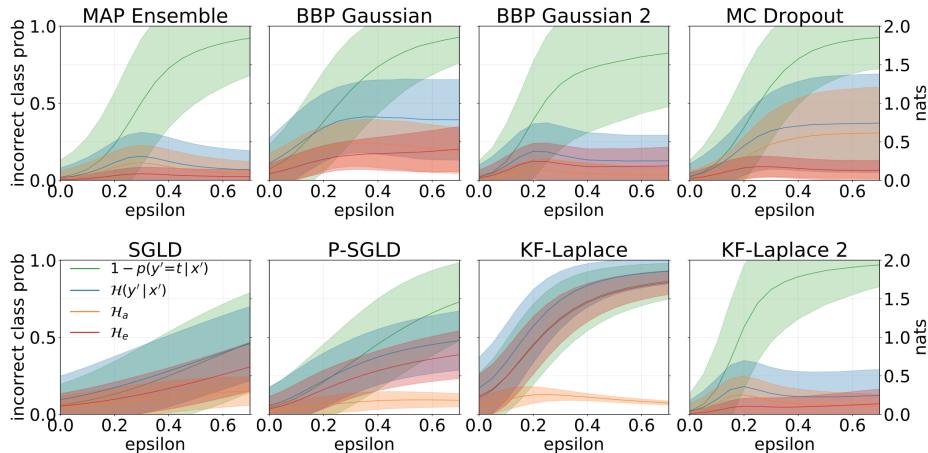
Deep learning models have implicitly redundant complexity.

Model complexity: application

Model	image size	# parameters	Mult-Adds	Top 1 Acc. (%)	Top 5 Acc. (%)
Inception V2 [29]	224×224	11.2 M	1.94 B	74.8	92.2
NASNet-A (5 @ 1538)	299×299	10.9 M	2.35 B	78.6	94.2
Inception V3 [59]	299×299	23.8 M	5.72 B	78.0	93.9
Xception [9]	299×299	22.8 M	8.38 B	79.0	94.5
Inception ResNet V2 [57]	299×299	55.8 M	13.2 B	80.4	95.3
NASNet-A (7 @ 1920)	299×299	22.6 M	4.93 B	80.8	95.3
ResNeXt-101 (64 x 4d) [67]	320×320	83.6 M	31.5 B	80.9	95.6
PolyNet [68]	331×331	92 M	34.7 B	81.3	95.8
DPN-131 [8]	320×320	79.5 M	32.0 B	81.5	95.8
SENet [25]	320×320	145.8 M	42.3 B	82.7	96.2
NASNet-A (6 @ 4032)	331×331	88.9 M	23.8 B	82.7	96.2

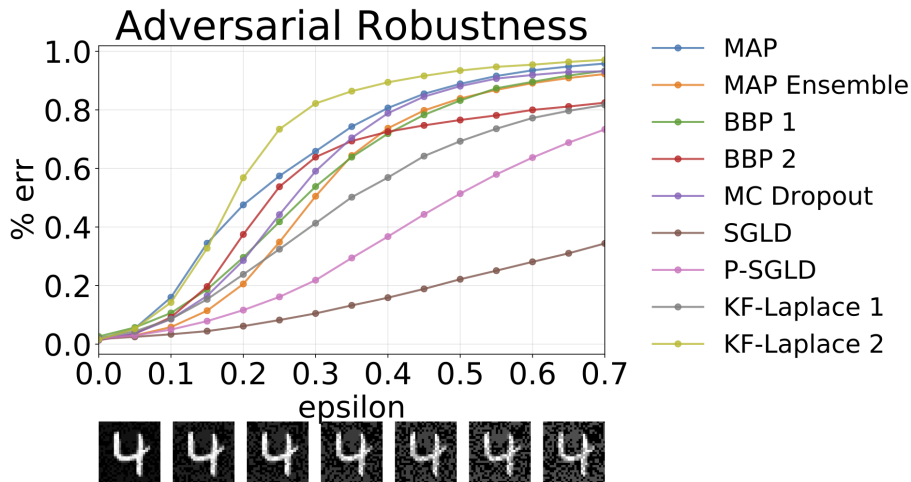
Zoph et al., 2017. Model parameter number differ twice, the performance is similar.

Robustness



<https://github.com/JavierAntoran/Bayesian-Neural-Networks>

Robustness



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Naive model complexity

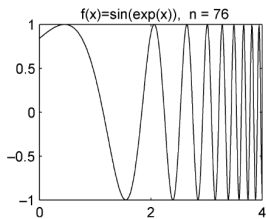
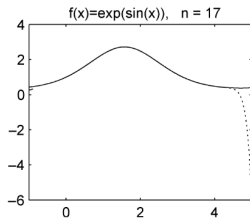
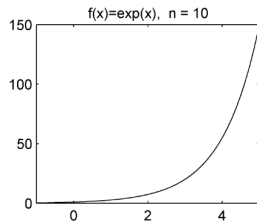
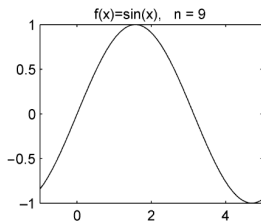
$C(\mathbf{f})$ is a number of parameters in the model.

Naive model complexity

$C(\mathbf{f}) = \text{number of parameters in the model}$

- Generalizes the idea of feature selection.
- Not differentiable.
- Ignores the model structure.

Geometric model complexity



Complexity as a regularizer

Complexity does not come alone: it is meaningful only together with a measure of how well the model explains the data.

- L2-regularization:

$$\|\mathbf{W}\|^2 - \log p(\mathbf{y}|\mathbf{X}, \mathbf{W}).$$

- Bayesian evidence:

$$\int_{\mathbf{W}} p(\mathbf{W}) p(\mathbf{y}|\mathbf{X}, \mathbf{W}) d\mathbf{W}.$$

Key idea: abstract complexity alone is not a useful model selection criterion — it must be related to the data.

Occam's razor



- William of Ockham: «Plurality must never be posited without necessity».

Occam's razor



- William of Ockham: «Plurality must never be posited without necessity».
- Modern interpretation: entities should not be multiplied beyond necessity.
- Paul Dirac: «A theory with mathematical beauty is more likely to be correct than an ugly one that fits some experimental data.»
- Albert Einstein: «Everything Should Be Made as Simple as Possible, But Not Simpler»

When Occam's razor does not necessarily work

Occam's razor is an empirical rule for sorting hypothesis during research.

It can be wrong:

Ernst Mach: molecules are a fictitious construct, as they are not observable.

Minimum description length

Task

Given a string: 001011001011001011... 001011, where the pattern 001011 repeats 100500 times.

How can we describe this string?

- `s == '001011...001011001011001011'`
- `s == ''.join('001011' for _ in range(100500))`

Kolmogorov complexity

Definition

Given a computable partially defined mapping from a set of binary words into itself:

$$T : \{0, 1\}^* \rightarrow \{0, 1\}^*.$$

Kolmogorov complexity of the binary string x is a minimal description length w.r.t. T :

$$K_T(x) = \min_{f \in \{0, 1\}^*} \{|f| : T(f) = x\},$$

Kolmogorov complexity

Generally, Kolmogorov complexity is uncomputable.

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Discrete distribution entropy

Definition

Given a discrete variable x with probability p and values $x_1, \dots, x_n$, An entropy of x is:

$$H(x) = - \sum_{i=1}^n p(x = x_i) \log p(x = x_i).$$

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- interpretation: measure of disorder in the distribution;
- maximal at uniform distribution;
- minimal at a distribution with concentration only at one event ($x_i = 1, x_j = 0, i \neq j$).

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- minimal at a distribution with concentration only at one event ($x_i = 1, x_j = 0, i \neq j$).
- **relation to Kolmogorov complexity:**

$$K(x) \leq \hat{H}(x) + O(\log n)$$

for binary string with length n .

Minimum description length principle

$$\text{MDL}(\mathbf{f}, \mathcal{D}) = L(\mathbf{f}) + L(\mathcal{D}|\mathbf{f}),$$

where $\mathbf{f}$ is a model, $\mathcal{D}$ is a dataset, L is a description length in bits.

$$\text{MDL}(\mathbf{f}, \mathcal{D}) \sim L(\mathbf{f}) + L(\mathbf{w}^*|\mathbf{f}) + L(\mathcal{D}|\mathbf{w}^*, \mathbf{f}),$$

$\mathbf{w}^*$ — optimal parameters.

$\mathbf{f}_1$	$L(\mathbf{f}_1)$	$L(\mathbf{w}_1^* \mathbf{f}_1)$	$L(\mathcal{D} \mathbf{w}_1^*, \mathbf{f}_1)$
$\mathbf{f}_2$	$L(\mathbf{f}_2)$	$L(\mathbf{w}_2^* \mathbf{f}_2)$	$L(\mathcal{D} \mathbf{w}_2^*, \mathbf{f}_2)$
$\mathbf{f}_3$	$L(\mathbf{f}_3)$	$L(\mathbf{w}_3^* \mathbf{f}_3)$	$L(\mathcal{D} \mathbf{w}_3^*, \mathbf{f}_3)$

MDL: example

Problem

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How can we describe this string?

- `s == '001011...001011001011001011'`
- `s == ''.join('001011' for _ in range(100500))`
- `import re; re.match('(001011){100500}')`

MDL: example

Problem

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How can we describe this string?

- `s == '001011...001011001011001011'`
- `s == ''.join('001011' for _ in range(100500))`
- `import re; re.match('(001011){100500}')`
- $L(\mathbf{f}_1) = 0, L(\mathcal{D}|\mathbf{f}_1) = 100505;$
- $L(\mathbf{f}_2) = 0, L(\mathcal{D}|\mathbf{f}_2) = 45;$
- $L(\mathbf{f}_3) \gg 0, L(\mathcal{D}|\mathbf{f}_3) = 38;$

MDL and Kolmogorov complexity

Kolmogorov complexity is the length of the shortest program that generates the dataset, written in a given programming language.

Invariance theorem

For any description language L there exists a constant c_L such that

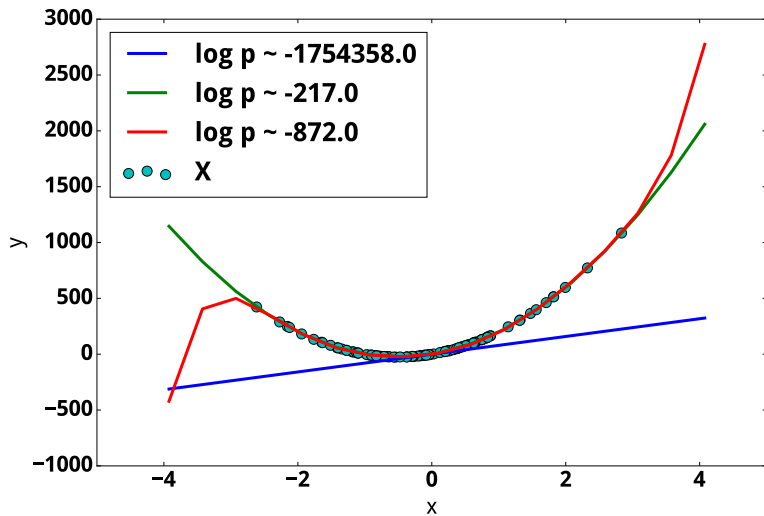
$$K_U(x) \leq K_L(x) + c_L,$$

where U is a fixed universal description language.

Difference from MDL:

- Incomputable.
- Code length depends on the language. For small sample size the invariance theorem can give poor results.

Probability coding



MDL + probability

Model selection problem can be viewed as a problem of information transmission, from encoder to decoder.

Given a dataset $\mathbf{X}, x \in \mathbf{X}$.

- Encoder encodes information about the dataset $\mathbf{X}$ with some code model f and transmits it to the decoder.
- Decoder decodes the information $f(\mathbf{X})$ and restores the data $\mathbf{X}$ (with loss).
- Problem is to find optimal coding method for x
- Code length: $-\log p(x)$

Quality criterion is *Regret*:

$$R(x) = -\log P(x) + \max_{f \in \mathfrak{F}} (\log P(x|f)).$$

It shows the difference between the length of the real code $-\log P(x)$ for x and the best code from the set of codes $\mathfrak{F}$.

Regret for datasets with parameterized distribution:

$$R(\mathfrak{X}) = \max_{\mathbf{X} \in \mathfrak{X}} (-\log P(\mathbf{X}) + \max_{\mathbf{w}} (\log P(\mathbf{X}|\mathbf{w}))).$$

MDL and Evidence

Statement

Let the likelihood function $p(\mathbf{X}|\mathbf{w}, \mathbf{f})$ be from an exponential family of distributions:

$$p(x|\mathbf{w}, \mathbf{f}) = h(x)g(\boldsymbol{\eta})\exp(\boldsymbol{\eta} \cdot \mathbf{T}(x)),$$

where $h, g, \mathbf{T}$ are some functions, $\boldsymbol{\eta}$ is a distribution parameter.

Let prior be Jeffreys prior:

$$p(\mathbf{w}|\mathbf{f}) = \frac{\sqrt{I(\mathbf{w})}}{\int_{\mathbf{w}} \sqrt{I(\mathbf{w})}},$$

Then regret and evidence differ only by some constant when $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \left(R(\mathbf{X}) - \log p(\mathbf{X}|\hat{\mathbf{w}}) - \log \int_{\mathbf{w}} p(\mathbf{X}|\mathbf{w}, \mathbf{f}) p(\mathbf{w}|\mathbf{f}) d\mathbf{w} \right) = \text{Const.}$$

Voita, Titov, 2020 ¹

Problem: evaluate how well language representations generalize to downstream probing tasks.

- Linguistic task: predict the linguistically correct part of speech.
- Control task: predict a randomly assigned part of speech.

A good representation allows the probing model to learn the linguistic structure quickly, resulting in a short code. For the control task, the labels have to be memorized, resulting in a longer code.

	Accuracy	Description Length			
		variational code		online code	
		codelength	compression	codelength	compression
MLP-2, h=1000					
LAYER 0	93.7 / 96.3	163 / 267	31.32 / 19.09	173 / 302	29.5 / 16.87
LAYER 1	97.5 / 91.9	85 / 470	59.76 / 10.85	96 / 515	53.06 / 9.89
LAYER 2	97.3 / 89.4	103 / 612	49.67 / 8.33	115 / 717	44.3 / 7.11

Experimental results; shown in pairs: linguistic task / control task. Codelength is measured in kbits

¹See also our project: https://github.com/intsystems/MDL_for_Language_Models

MDL vs. Evidence

Evidence	MDL
Uses prior knowledge	Uses a coding scheme
Uses data generation hypothesis	Minimizes description length

Solomonoff induction ²

A combination of Bayesian inference and MDL using Kolmogorov complexity (!) for programs.

- Hypotheses = programs.
- Shorter programs x receive larger prior weight:

$$p(x) = 2^{-|x|}.$$

- The Solomonoff prior is a mixture over all programs:

$$p(y) = \sum_{x:U(x)=y*} 2^{-|x|}.$$

- This mixture is closely related to Kolmogorov complexity:

$$-\log p(y) = K(y) + O(1).$$

- In Bayesian terms, this is a form of Bayesian model averaging over programs.

²See the talk of Vadim Kasiuk, 2025

Recap: Likelihood maximization and KL divergence

Suppose the true data distribution is $p^*(x)$ and our model is $p(x | w)$.

Then

$$KL(p^*(x) || p(x | w)) = \mathbb{E}_{x \sim p^*} \left[\log \frac{p^*(x)}{p(x | w)} \right].$$

Therefore

$$KL(p^* || p(\cdot | w)) = \text{Const} - \mathbb{E}_{x \sim p^*} \log p(x | w).$$

By the law of large numbers,

$$\mathbb{E}_{x \sim p^*} \log p(x | w) \approx \frac{1}{N} \sum_{i=1}^N \log p(x_i | w).$$

Hence, asymptotically,

$\text{MLE} \equiv \text{KL minimization}$

under the usual regularity assumptions.

Akaike information criterion (AIC)

Main idea

- We want to minimize the KL divergence between the true distribution and the fitted model:

$$KL(p^*(\mathbf{X}) || p(\mathbf{X}|\mathbf{w})).$$

- Equivalently, maximize the expected log-likelihood:

$$\mathbb{E}_{p^*} \log p(\mathbf{X}|\mathbf{w}).$$

- A naive estimation $\mathbb{E} \log p(\mathbf{X}|\mathbf{w}) \approx \log p(\mathbf{X}|\mathbf{w})$ is optimistically biased estimate of the expected test log-likelihood.
- AIC corrects this bias:

$$AIC = -2 \log p(\mathbf{X}|\hat{\mathbf{w}}) + 2k,$$

where $k = \dim(\mathbf{w})$.

Bayesian information criterion (BIC)

Main idea

- Consider evidence with flat prior:

$$p(\mathbf{X}) = \int p(\mathbf{X}|\mathbf{w})p(\mathbf{w}) d\mathbf{w}.$$

- Use Laplace approximation around the MLE:

$$\log p(\mathbf{X}) \approx \log p(\mathbf{X}|\hat{\mathbf{w}}) - \frac{k}{2} \log n + O(1),$$

where $k = \dim(\mathbf{w})$.

- This gives the BIC:

$$BIC = -2 \log p(\mathbf{X}|\hat{\mathbf{w}}) + k \log n.$$

AIC vs BIC

- $BIC > AIC$ when $n \geq 8$
- Model selection with BIC leads to consistent estimation: when n grows, a probability of correct model selection converges to 1
- AIC minimization asymptotically gives the model with maximal likelihood
- Heuristic: models, which criteria values differ by 1 or 2 from best values are distinguishable.
- Both criteria work poorly for large models ³.

³See also a talk on WBIC by Ernest Nasyrov

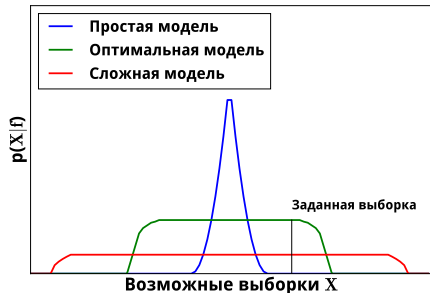
Cross-validation as a model selection method

Pros

- Easy to implement.
- Straightforward implementation is simple for understanding. No approximations, no tricks.
- Makes few assumptions about the model.

Cons

- Either we lose data part, or we run performance.
- Does not explicitly characterize the parameter-space volume around the optimum.



Evidence vs Cross-validation

Evidence:

$$\log p(\mathbf{X}|\mathbf{f}) = \log p(\mathbf{x}_1|\mathbf{f}) + \log p(\mathbf{x}_2|\mathbf{x}_1, \mathbf{f}) + \cdots + \log p(\mathbf{x}_n|\mathbf{x}_1, \dots, \mathbf{x}_{n-1}, \mathbf{f}).$$

Leave-one-out:

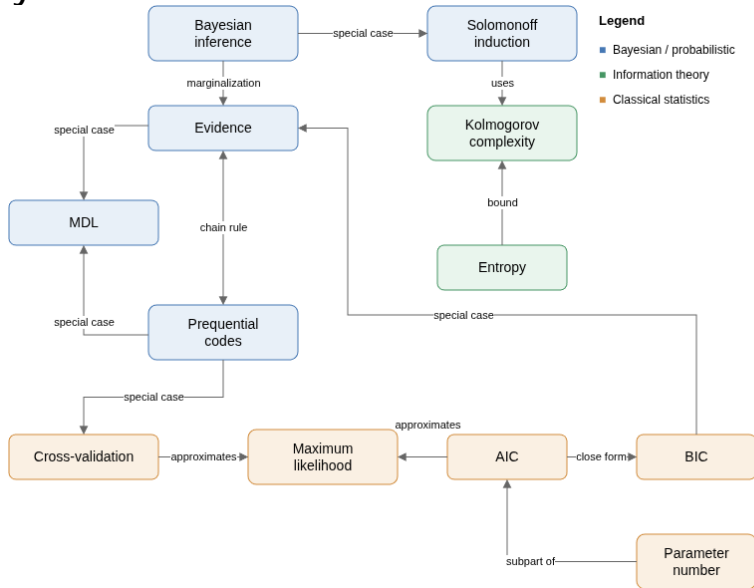
$$\text{LOU} = E \log p(\mathbf{x}_n|\mathbf{x}_1, \dots, \mathbf{x}_{n-1}, \mathbf{f}).$$

Cross-validation uses average value of the last term $p(\mathbf{x}_n|\mathbf{x}_1, \dots, \mathbf{x}_{n-1}, \mathbf{f})$.

Evidence considers **full** description length.

For a fixed probabilistic model, the prequential decomposition of the joint likelihood is exactly the evidence code. Practical prequential MDL (or online code) additionally refits the model as more data become available.

Complexity and friends



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